

# Speed Control of Low-Speed PMSM Based on improved Proportional-Resonant and Kalman Filter Strategy

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**Abstract**—Permanent magnet synchronous machines (PMSMs) suffer from degraded speed control accuracy due to periodic disturbances and insufficient position sensor resolution under low-speed operation. First, the improved proportional-resonant (PR) controller is proposed to enhance the capability to suppress periodic disturbances with unknown frequencies by parallel connecting an adaptive quasi-resonant controller. Furthermore, a dynamic position estimation method based on Kalman filter is employed to mitigate position sampling errors caused by low-resolution position sensors. Experimental results demonstrate that the proposed strategy significantly enhances system operation stability and markedly improves speed control precision

**Keywords**—improved proportional-resonant controller, periodic disturbances, Kalman filter

## I. INTRODUCTION

Low-speed PMSMs play critical roles in high-precision applications such as medical equipment, astronomical observation, and aerospace systems. The speed control accuracy fundamentally relies on precise position information and advanced speed control strategies. However, the low-speed operation is affected by periodic disturbances such as current sampling errors, flux harmonics, and cogging torque. Furthermore, the insufficient resolution of position sensors at low speed such as 1 r/min tends to amplify position sampling errors and significantly degrades speed control accuracy.

For position estimation, conventional rotor position estimation strategies were presented, including interpolation methods, synchronous coordinate filter, and position vector observers [1]. However, the average acceleration method exhibits significant latency during low-speed operation. Additionally, the back- electromotive force (BEMF) based speed observer method, which is employed in sensorless control, similarly suffers from poor estimation accuracy under low-speed operation. Therefore, it is essential to

investigate a position estimation method for low-speed machine.

To address periodic disturbances, a repetitive control strategy was proposed to address the harmonic disturbances [2]. In [3], an improved active disturbance rejection controller (ADRC) was proposed to suppress periodic disturbances. In [4], the influence of periodic cogging torque disturbances is considered, and a virtual cogging torque control method is proposed to reduce the impact of speed fluctuations. However, these strategies are limited to the suppression of known periodic disturbances. During the operation of low-speed machine, variations in machine parameters and load may induce uncertain periodic disturbances, making it necessary to develop a strategy for suppressing periodic disturbances with unknown frequencies.

This paper proposes a speed control strategy for low-speed PMSM drive. First, an improved PR controller is designed to suppress periodic disturbances with unknown frequencies, significantly migrates the steady-state speed fluctuation. Additionally, to address the position sampling error associated with low-resolution position sensors, a dynamic position estimation method based on a Kalman filter is employed to enhance the speed control accuracy.

## II. ANALYSIS OF PERIODIC DISTURBANCES AND POSITION SAMPLING ERRORS IN LOW-SPEED PMSM

### A. Analysis of Periodic Disturbances

Periodic disturbances in low-speed PMSMs primarily originate from current sampling errors, flux harmonics, and cogging torque.

(1) Current sampling errors induce 1st and 2nd harmonic torque components. The mathematical representation of this disturbance is:

$$T_{offset} = \frac{2}{\sqrt{3}} \cos(\theta_e + \alpha) \sqrt{\Delta i_a^2 + \Delta i_a \Delta i_b + \Delta i_b^2} \quad (1)$$

where  $\Delta i_a$  and  $\Delta i_b$  denote the DC offset errors caused by sampling circuits, and  $\alpha$  is a constant angle. The torque harmonics caused by the current scaling error can be

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expressed as:

$$T_{scaling} = \frac{I_m}{\sqrt{3}} \frac{K_a - K_b}{K_a K_b} \cos \left[ (2\theta_e + \frac{\pi}{3}) + \frac{1}{2} \right] \quad (2)$$

where  $I_m$  is the amplitude of phases current,  $K_a$  and  $K_b$  denote the actual-to-measured current gains for phases a and b, respectively.

(2) Flux harmonics generate 6th harmonic torque components. The induced flux harmonics can be mathematically expressed as:

$$\begin{cases} T_{\psi d} = \sum_{n=1}^{\infty} K_d \cos(6n\theta_e) \\ T_{\psi q} = \sum_{n=1}^{\infty} K_q \sin(6n\theta_e) \end{cases} \quad (3)$$

where  $T_{\psi d}$  and  $T_{\psi q}$  are the flux harmonics in the d and q-axis frames, respectively.  $K_d$  and  $K_q$  denote the amplitudes of the flux harmonics. As a result, flux harmonics are predominantly manifested as disturbance components located at the 6 $n$ th harmonic orders.

(3) The cogging torque produces torque ripple with a periodicity equal to the least common multiple of the number of slots and the number of pole pairs. The mathematical representation is expressed as:

$$T_{cog} = -\frac{\pi N_z L_{Fe} (R_s^2 - R_r^2)}{4\mu} \sum_{n=1}^{\infty} n \sin(nN_{pz} \theta_e) \quad (4)$$

where  $N_z$  is the number of slots,  $L_{Fe}$  is the length of the iron core,  $R_s$  and  $R_r$  represent the inner radius of the stator and the outer radius of the rotor, respectively.  $\mu$  is the air-gap permeability,  $N_{pz}$  is the least common multiple of the machine's slot number and pole pairs and  $\theta_e$  is the rotor electrical angle.

In addition, factors such as variations in machine parameters and load conditions may induce uncertain disturbances and it is difficult to determine its exact expression.

### B. Analysis of Position Sampling Error

Resolvers are widely used in position detection for servo systems owing to their excellent environmental adaptability and cost-effectiveness. In this experiment, a single-pair absolute resolver with a 16-bit resolution resolver-to-digital converter (RDC) is employed to analyze the position sampling errors under the operating condition of a low-speed PMSM at 1 r/min. The analysis is performed as follows, with a sampling frequency of 10 kHz.

$$\Delta\theta_{min} = \frac{2\pi}{2^{16}} = 9.58 \times 10^{-5} \text{ rad/s} \quad (5)$$

$$\Delta\theta_{run} = \frac{2\pi}{60 \times 10000} = 1.04 \times 10^{-5} \text{ rad/s} \quad (6)$$

where  $\Delta\theta_{min}$  denotes the minimum resolvable angle per sampling cycle,  $\Delta\theta_{run}$  represents the actual rotational angle of the machine. The inequality  $\Delta\theta_{min} > \Delta\theta_{run}$  indicates that under low-speed operation at 1 r/min, the rotor position estimation is subject to quantization errors, resulting in inaccurate speed that fails to reflect the true machine dynamics.

To validate the impact of position sampling errors on control performance, a simulation model of the low-speed PMSM system is developed. Comparative experiments are

conducted with the machine operating at 1 r/min. Rotor positions are discretized at a 10 kHz sampling frequency to emulate position sampling errors, as illustrated in Fig. 1 and the simulation results of speed by PI control are shown in Fig. 2.

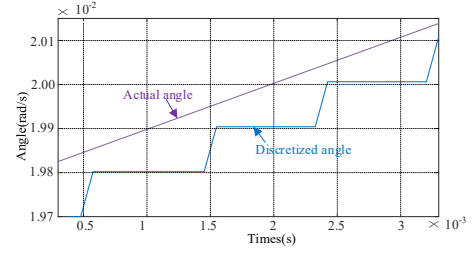


Fig. 1 Diagram of discretized rotor angle

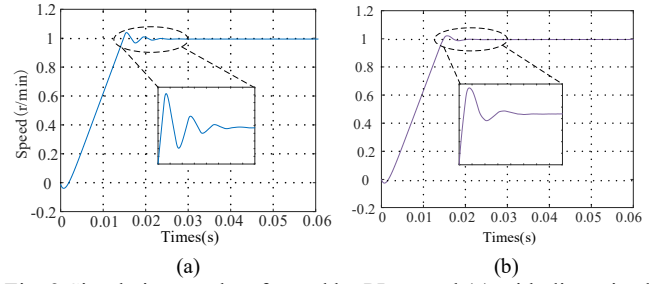


Fig. 2 Simulation results of speed by PI control (a) with discretized angle. (b) with actual angle.

From Fig.2, the discretized angle degrades control performance: the speed settling time increased by 75% from 0.02 s to 0.035 s. Furthermore, the speed overshoot increased from 0.015 r/min to 0.02 r/min, corresponding to a 33% reduction in dynamic performance. Therefore, it is necessary to develop a position estimation method to solve this problem.

## III. LOW-SPEED PMSM CONTROL STRATEGY

### A. Improved PR Controller

To suppress periodic disturbances with unknown frequencies, this paper proposed an improved PR controller. Conventional PR controller suffers from limited stability margins and poor dynamic response. A quasi-resonant (QR) controller is widely used, which is expressed as:

$$G_{qr}(s) = \frac{2K_1\omega_c s}{s^2 + 2\omega_c s + \omega_0^2} \quad (7)$$

where  $K_1$  denotes the resonant gain,  $\omega_c$  denotes the resonant bandwidth, and  $\omega_0$  denotes the resonant frequency.

During the operation of a low-speed machine, factors such as variations in motor parameters and load changes can induce unknown periodic disturbances. To address this issue, an improved PR controller has been proposed, as illustrated in Fig. 3.

As shown in Fig. 3, the quasi-resonant controller is connected in parallel with an adaptive quasi-resonant controller. This approach generates an adaptive resonant frequency  $\omega_1$  through PI regulation based on the speed error and output of the improved PR controller, thereby achieving suppression of the unknown periodic disturbances. The transfer function of this improved PR controller is:

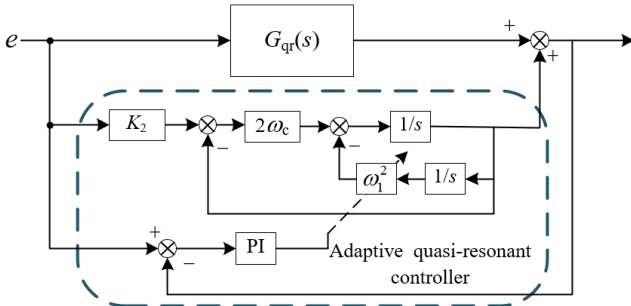


Fig. 3 Diagram of improved PR controller.

$$G_c(s) = G_{qr}(s) + \frac{2K_2\omega_c s}{s^2 + 2\omega_c s + \omega_l^2} \quad (8)$$

The simplified block diagram of speed loop is shown in Figure 4, where  $G_c(s)$  represents the transfer function of the proposed improved PR controller,  $T_d$  represents the periodic disturbance torque, and  $\omega_i$  represents the current loop bandwidth.

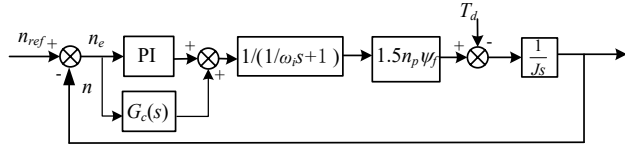


Fig. 4 Simplified block diagram of speed loop.

The open-loop transfer function of the speed loop can be expressed as follows:

$$G_0(s) = \frac{45G_c(s)n_p\psi_f}{\pi J(1/\omega_i s^2 + s)} \quad (9)$$

To investigate the disturbance rejection characteristics of the improved PR controller, the periodic disturbance  $T_d$  is set as the input signal, while the speed error  $n_e$  serves as the output. The block diagram of disturbance loop is illustrated in Figure 5.

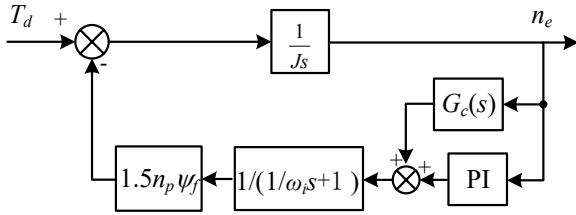


Fig. 5 Block diagram of disturbance loop.

The closed-loop transfer function between the speed error and the periodic disturbance can be expressed as:

$$G_{Td}(s) = \frac{30(1 + 1/\omega_i s)}{\pi J s(1 + 1/\omega_i s) + 45n_p\psi_f G_c(s)} \quad (10)$$

From (9), the open-loop frequency response characteristics of the speed loop with the PI, conventional PR and improved PR controller are shown in Fig. 6(a). From Fig. 6(a) the proposed improved PR controller exhibits increased gain at the resonant frequency compared to PI controller, as well as enhanced the stability margin compared to conventional PR controller.

From (10), Fig. 6(b) demonstrates its disturbance rejection characteristics. The proposed improved PR controller

significantly enhances disturbance rejection capabilities in the low frequency range compared to conventional PR controller and enhance the performance of disturbance rejection compared to PI controller.

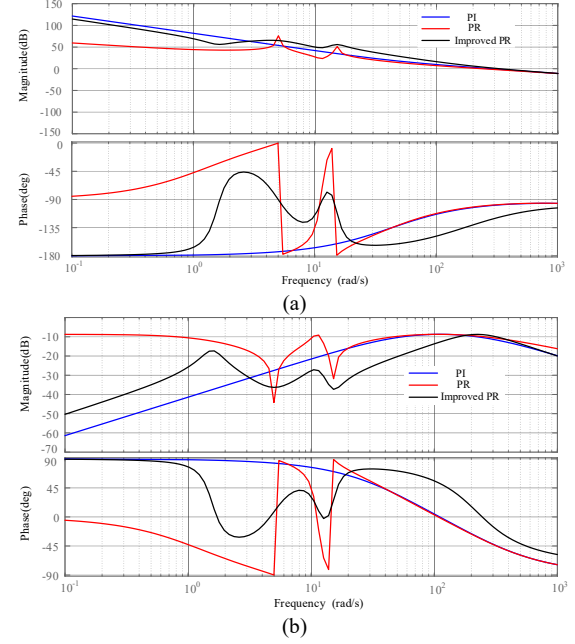


Fig. 6 Bode diagrams of improved PR controller

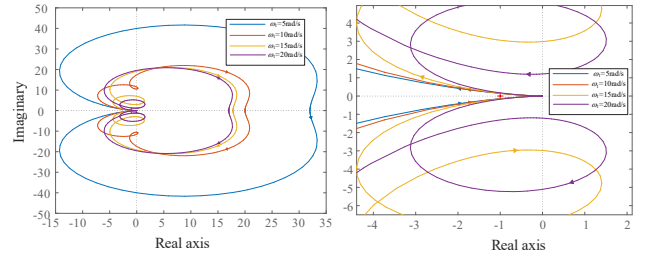


Fig. 7 Nyquist plots for different resonant frequencies

Meanwhile, Fig. 7 illustrates the Nyquist plot of the improved PR controller. When the resonant frequency varies within the range of 5 to 20 rad/s, all open-loop poles remain in the left-half plane, and the Nyquist curve does not encircle the  $(-1, j0)$  point, thereby validating the stability of the proposed strategy by Nyquist stability theory.

### B. Position Estimation Strategy Based on Kalman Filter

Accurate rotor position estimation is critical for low-speed PMSM control. However, the 16-bit resolution RDC used in this experiment induced significant position sampling errors. To solve this problem, this paper integrates a Kalman filter into a speed-acceleration state observation equation, enabling dynamic position estimation.

The Kalman filter is a state estimation method based on a recursive algorithm that enables optimal estimation of dynamic systems from noisy observation sequences. It comprises two stages: prediction and update, which correspond to (11) and (12), respectively:

$$\begin{cases} \hat{x}_t^- = A\hat{x}_{t-1} + Bu_{t-1} \\ P_t^- = AP_{t-1}A^T + Q \end{cases} \quad (11)$$

$$\begin{cases} K_t = P_t^- H^T (H P_t^- H^T + R)^{-1} \\ \hat{x}_t = \hat{x}_t^- + K_t (z_t - H \hat{x}_t^-) \\ P_t = (I - K_t H) P_t^- \end{cases} \quad (12)$$

The prediction phase computes the priori estimate  $\hat{x}_t^-$  and covariance matrix  $P_t^-$ , by the state observation equation, while the update phase refines the posteriori estimate  $\hat{x}_t$  and covariance  $P_t$  by integrating the measurement  $z_t$ . Here,  $K_t$  denotes the Kalman gain.  $Q$  and  $R$  represent the process noise covariance and measurement noise covariance matrices, respectively.  $H$  is the observation matrix;  $A$  and  $B$  are the state transition matrices;  $u_t$  corresponds to the system input variable.

The Kalman filter-based state observation equation combined machine speed and acceleration to achieve dynamic position estimation. The augmented state vector is defined as  $x=[\omega_a, \omega_b]^T$ , where  $\omega_a$  denotes the machine speed and  $\omega_b$  represents the speed fluctuations during operation. Based on the machine's mechanical dynamics, the Kalman filter state observation equation is formulated as follows, where acceleration  $a$  is derived from the speed reference.

$$\begin{bmatrix} \hat{\omega}_{at}^- \\ \hat{\omega}_{bt}^- \end{bmatrix} = \begin{bmatrix} 1 & -\Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{\omega}_{at-1}^- \\ \hat{\omega}_{bt-1}^- \end{bmatrix} + \begin{bmatrix} \Delta t \\ 0 \end{bmatrix} a \quad (13)$$

In addition, the process noise, measurement noise, and initial covariance values  $Q$ ,  $R$ , and  $P_0$  were calibrated experimentally as:

$$\begin{cases} Q = \begin{bmatrix} 0.0001 & 0 \\ 0 & 0.0001 \end{bmatrix} \\ R = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \\ P_0 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \end{cases} \quad (14)$$

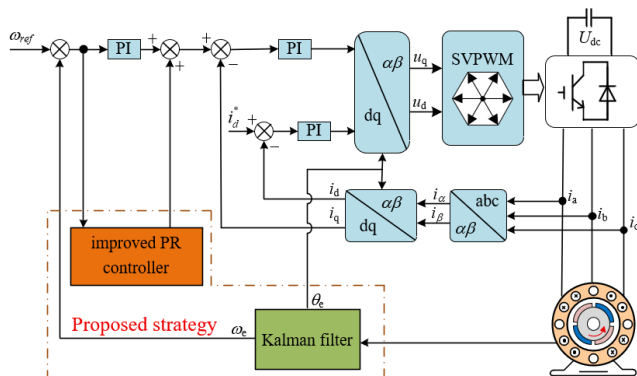


Fig. 8 The diagram of proposed strategy

In summary, the diagram of the proposed strategy is illustrated in Fig.8. From Fig. 8, the proposed position method enhances the accuracy of speed sampling so that the proposed improved PR controller can effectively suppress the periodic disturbances and migrate speed fluctuations.

## IV. EXPERIMENTAL VERIFICATION

The low-speed PMSM test platform is shown in Fig. 8. The load machine is driven by a separate controller and the system

parameters are provided in Table I.

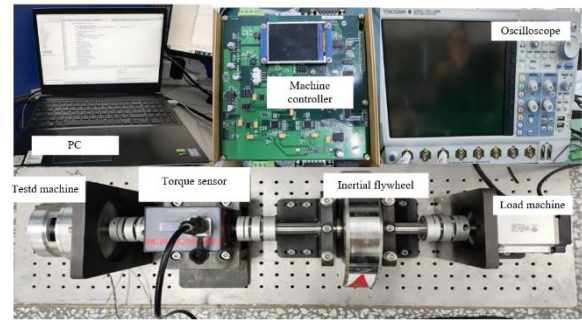


Fig.8 low-speed PMSM experimental platform

TABLE I  
THE PARAMETERS OF LOW-SPEED PMSM

| Parameters           | Symbol   | Value                   |
|----------------------|----------|-------------------------|
| Number of pole pairs | $p_n$    | 8                       |
| Rated torque         | $T_n$    | 0.4 N·m                 |
| Rated current        | $I_n$    | 1.2 A                   |
| Inductance           | $L_s$    | 10 mH                   |
| Stator resistance    | $R$      | 10 $\Omega$             |
| Machine inertia      | $J$      | 0.093 kg·m <sup>2</sup> |
| Flux linkage         | $\psi_f$ | 0.05 Wb                 |
| Number of slots      | $N_z$    | 18                      |
| DC-link voltage      | $U_{dc}$ | 30 V                    |

As analyzed previously, the position sampling error induced by the low-resolution RDC significantly impacts the speed control accuracy of low-speed machines. The experimental results are shown in Fig. 9 and the speed reference was set as 1 r/min. From Fig. 9(a) position sampling error induced speed fluctuations with an amplitude of 2 r/min, while the proposed method based on Kalman filter achieved further suppression to 0.40 r/min, corresponding to a 78.5% reduction.

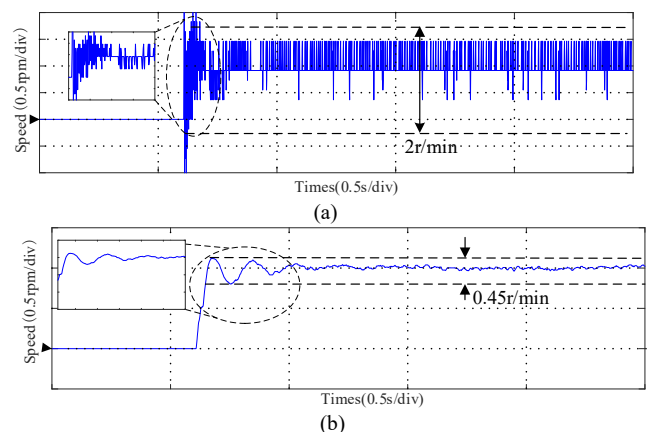


Fig. 9 The experimental results of 1 r/min. (a) without any suppressions. (b) with proposed method based on Kalman filter.

To further verify the dynamic performance of the proposed method. The speed reference was set as  $\sin(10t)$  r/min and the experimental results are shown in Fig. 10. The results indicate



that the proposed method reduced the amplitude of the speed error from 2.5 r/min to 0.5 r/min, corresponding to a suppression rate of 80%.

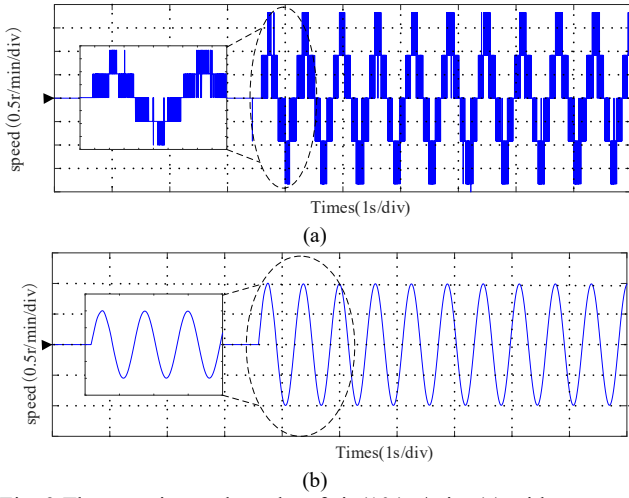


Fig. 9 The experimental results of  $\sin(10t)$  r/min. (a) without any suppressions. (b) with proposed method based on Kalman filter.

To verify the effectiveness of the proposed improved PR controller, comparative tests were conducted by PI, conventional PR and improved PR controller on the basis of position estimation based on Kalman filter. The speed reference was set as 1 r/min and the load was set as rated value. The steady-state speed performances are shown in Fig. 10.

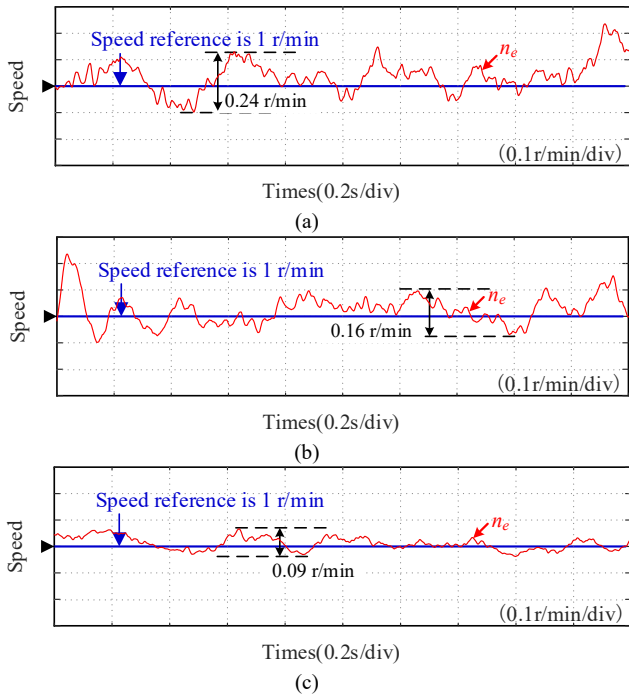


Fig. 10 The experimental results of steady-state speed under 100% rated load. (a) PI. (b) conventional PR controller. (c) proposed improved PR controller.

From Fig. 10 the steady-state speed fluctuation of PI is 0.24 r/min. The conventional PR reduced to 0.16 r/min, while the proposed improved PR reduced to 0.09r/min, corresponding to a suppression rate of 67.5%. Because the proposed improved PR controller enhances the capability to suppress

periodic disturbances with unknown frequencies compared to conventional PR controller. Therefore, the proposed improved PR controller can effectively suppress the periodic disturbances.

## V. CONCLUSIONS

The paper proposed a speed control strategy for low-speed PMSM by integrating an improved PR controller with a position estimation method based on Kalman filter. This approach aims to solve the problem of coupling effects between periodic disturbances with unknown frequencies and position sampling errors caused by low-resolution RDC. The experimental results demonstrate that the proposed position estimation method based on Kalman filter achieved enhanced precision in speed control and excellent dynamic performance. Furthermore, the proposed improved PR controller significantly migrates the steady-state speed fluctuation and effectively suppress periodic disturbances.

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